

Iranian, Armenian, and contact-induced syntactic change: *a status quaestionis*

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1 Background

The Armenian people and their language are traditionally said to have lived and developed in Greater Armenia, that is the Armenian Highlands. This area is roughly delimited by the Caucasus in the North, Media Atropatene (modern Azerbaijan) to the East, the Taurus Mountains to the South and the Upper Euphrates in the West.

The name Armenia is first attested as a geo-political term in the inscription of Darius at Bisotun (DB I.15 etc.); during his rule, Armenia was an Achaemenid satrapy. Although its political status, (in-)dependence from other powers, and borders should change multiple times over the coming millennium, Armenia was ruled by a succession of Iranian dynasties until the Arab Conquest in c. 645 CE. Figures 1 and 2 illustrate the changing borders, showing their greatest extent in the middle of the 1st century BCE under King Tigranes II, the Great, and in c. 379 CE at the death of the Sasanian King Šāpuhr II.

The Armenian language was thought to be a member of the Iranian branch of Indo-European languages until a seminal paper by Heinrich HÜBSCHMANN in 1875. With the realisation that the Iranian traits of Armenian were not inherited but borrowed, a new strand of research activity in Armenian studies commenced. Antoine MEILLET was the first to suggest that the model language for most Iranian loanwords in Armenian is Parthian (1911–1912). In the course of the 20th century, one of the most productive contributors to the understanding of this contact situation was Giancarlo BOLOGNESI, who proposed, amongst many other things, a stratification of Iranian loans (Old Iranian, Parthian phase I and II, Middle Persian) on the basis of various phonological features of such borrowings (1951; 1960).

As with the majority of language contact situations, lexical, phonological, morphological, and to a certain extent phraseological loans have been discussed in great detail over the course of the last century; for an overview, see MEYER (in pressa); SCHMITT (1983). One aspect of language contact that has thus far been largely ignored is syntax.

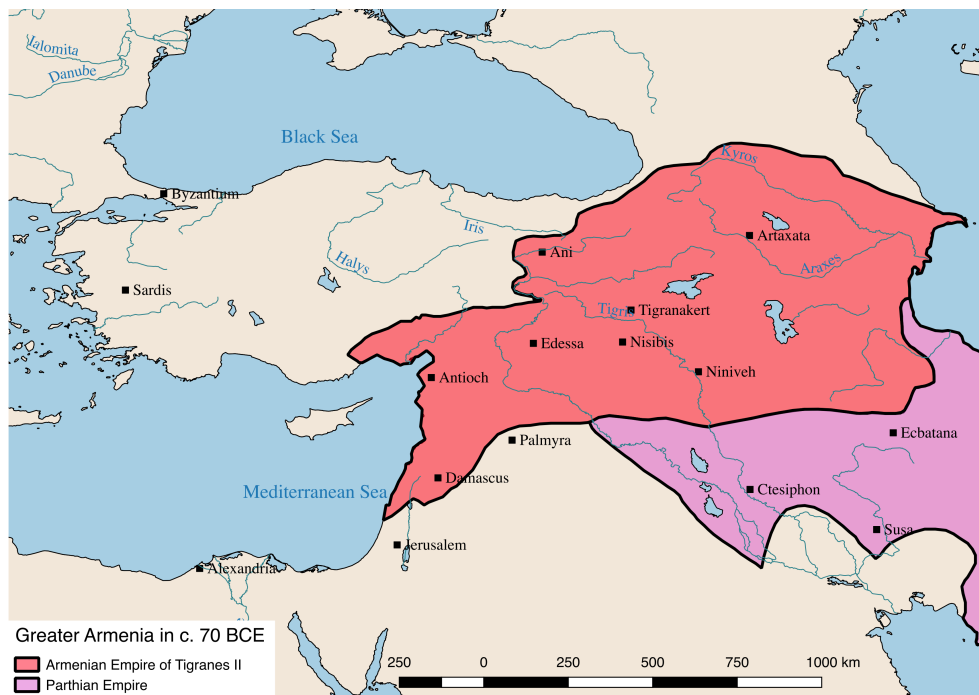


Figure 1: Map of Armenia in c. 70 BCE

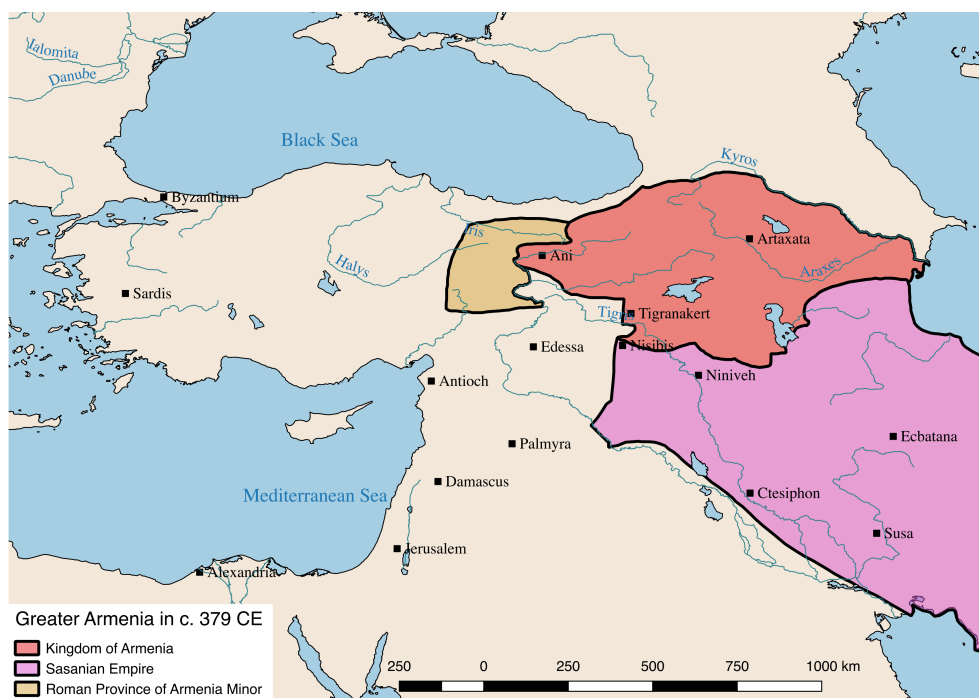


Figure 2: Map of Armenia in c. 379 CE

In what follows, a brief overview of the lexical, phonological, morphological, and to a certain extent phraseological borrowings will be provided (§2). In §3, three less complex syntactic patterns are discussed briefly, all of which have very close Iranian parallels. A more complex pattern, viz. the periphrastic perfect of Classical Armenian, is discussed in more detail in §4. §5 takes the linguistic data presented, and together with socio-historical data gleaned from literature briefly suggests a likely language contact scenario under which Armenian developed.

2 Iranian borrowings

2.1 Old Iranian

The evidence for Old Iranian words in Armenian is relatively scant; morphological and phonological peculiarities do, however, point to a pre-Middle Iranian, and thus most likely Achaemenid Old Persian origin for a small number of lexical items.

Attested in OP *ariya-*, Av. *airiia-* ‘Aryan’, this self-designation of the Iranians conspicuously has been borrowed into Armenian twice, first as *Arik*‘ and later as *Eran* (< MP ‘yr’n /Ērān/). Owing to its GEN.PL *Areac*‘, an *-ea-* stem like *teli*, *teleac*‘ ‘place’, the former is deemed to be a reflex of the Old Persian form, where Arm. *-ea-* reflects the Iranian *-ya-* stem (cf. MEILLET 1911–1912:249–50; SCHMITT 1983:77; DE LAMBERTERIE 1989). A similar age is envisaged for Arm. *tʻšnami* ‘enemy, hostile’, cp. Av. *dušmainiiu-* ‘id.’ (cf. SCHMITT 1980:422–3).

The culturally significant term Arm. *partēz* ‘garden, park; paradise’ is of particular interest for relative chronology; if indeed borrowed in Achaemenid times, the word must have undergone an inner-Armenian sound-change **-d-> -t-* (cp. Av. *pairidaēza-* ‘fenced area’, Gk. *παράδεισος*), but clearly not **p-> pʻ- / h- / Ø*.

Some scholars suggest further loans from (Middle) Median (FRYE 1969; PÉRIKHANIAN 1966); as pointed out by SCHMITT (1983), such derivations are difficult to verify owing to the paucity of Median data.

2.2 Parthian

2.2.1 Phonemic correspondences

Parthian has without doubt exerted the most significant influence on Armenian, both in terms of quantity of loan material as well as quality, viz. types of loans; in view of the almost 400 years that Armenia was under immediate Parthian rule, this is hardly surprising. In turn, the Armenian *Nebenüberlieferung* has proved to be instrumental for the reconstruction of not otherwise attested Middle Iranian data.

SCHMITT (1983:80–1, 96–7) provides an overview delineating both the dialectal differences between North West Parthian and South West Middle Persian forms, and the correspondence of Armenian and Parthian sounds. As a result of the different phonemic inventories of both languages, this correspondence is not biunique, but reasonably straightforward nonetheless. With the exception of Arm. *c*, *č*‘, and *c*‘ all Armenian sounds have at least one correspondence in Parthian, e.g. Arm. *b* < Pth. *b*. Where Armenian does not have a sound corresponding to that of Parthian, it is merged with a close approximation, so for example Arm. *g* < Pth. *g*, *γ*. Some examples include: Arm. *xoyr* ‘headgear, diadem’ < Pth. *xwwd* /xōδ/ ‘helmet’; Arm. *hraman* ‘command’ < WMIr. *frm’n* /framān/; Arm. *aržan* ‘worthy, proper’ < Pth. *ʻrj’n* /aržān/; Arm. *bžišk* ‘doctor, physician’ < WMIr. *bzyšk* /bizešk/.

While a large number of Armenian phonemes occur in both the inherited lexicon and in Iranian

loanwords, some consonants and consonant clusters are indicative of non-indigenous material; these include Arm. *p* and *č*, and in many instances *š*, *ž*, and *x*. Similarly symptomatic are the clusters Arm. *hr* < WMIr. *fr*, and with metathesis Arm. *šx* and *rh* < WMIr. *xš* and *hr*. Examples include: Arm. *ašxat* ‘trouble, labour’ < Pth. **xš’dyft* /ašxādift/ ‘wretchedness’.

2.2.2 Stratification

Correspondences in the vowel system are unproblematic for the most part in showing no changes; a great number of them also applies to loans from Middle Persian. It is possible, however, to differentiate two chronological strata in the material modelled on Parthian on the basis of three sets of vocalic correspondences (cf. BOLOGNESI 1951 and Table 1).

One set of loans show a correspondence Pth. *ō, ē* Arm. *oy, ē* (tonic) and *u, i* (pre-tonic): Arm. *boyž* ‘cure, remedy’, *bužem* ‘to cure, heal’ < Pth. *bwj-* /bōž-/ ‘to save, redeem’; Arm. *dēm̄k* ‘face’ < WMIr. *dym* /dēm/, Arm. *spitak* ‘white’, cp. Pth. **spyd* /ispēd/, MP *spyt* /spēd/. As both sounds undergo ablaut alternation within Armenian (*dēm̄k*, GEN *dimac*), and coincide with the outcome of the PIE diphthongs **eu*, **ou* > PArm. **ou* > Arm. *oy* and **oi*, **ei* > PArm. **ei* > Arm. *ē*, the Parthian loans in question must date to a time where Pth. *ō, ē* were still true diphthongs and close to the Proto-Armenian stage described.

The second set of loans, by contrast, yields Arm. *o, e* < Pth. *ō, ē*, and shows no differentiation between tonic or pre-tonic position: Arm. *den*, GEN *deni* ‘religion, faith’ < Pth. *dyn* /dēn/, Arm. *hreštak* ‘angel, messenger’ < Pth. *fryšt̄g* /frēšt̄g/; Arm. *rot* ‘river’ < WMIr. *rwd* /rōd/, Arm. *tohm* ‘family, seed’, cp. Pth. *twxm* /tōxm/, MP *twhm* /tōhm/. These developments suggest that at the time these words were replicated in Armenian, ablaut alternation was no longer productive, and the former Parthian diphthongs had completely monophthongised.

This stratal differentiation finds further expression in the rendition of Pth. *r-* in Armenian. While the chronologically earlier layer of borrowings shows prothetic vowels prepended to *r-*, the later layer uses Arm. *ř-* instead: Arm. *eram* ‘troop, flock’ < WMIr. *rm* /ram/ ‘flock; Manichaean community’; Arm. *razm* ‘fight, battle’ < WMIr. *rz̄m* /razm/.

Arm. (early loans)	Arm. (later loans)	Pth.
oý, u	o	w /ō/
é, i	e	y /ē/
er-	ř-	r- /r-/

Table 1: Stratal differentiation of loans from Parthian

BOLOGNESI (1960) dates both strata date to the Arsacid period; this suggestion is corroborated by the lexical nature of the loans. An earlier import cannot be excluded in principle, but is unlikely for the very same reason.

2.2.3 Types of lexical loans

The features that make Parthian loanwords in Armenian stand out are their categorial pervasiveness and general abundance; Parthian material is not restricted to any part of the lexicon, or indeed any one grammatical category, but is found across the spectrum in both the basic lexicon (items concerning nature, body parts, abstract vocabulary of everyday life, etc.) and in more specialised segments (e.g.

martial and technical vocabulary), in both of which they may occur as nouns, adjectives, adverbs, verbs, even invading closed classes such as prepositions and numbers.

Examples of such less common borrowings are few in number, but significant nonetheless. The preposition Arm. *vasn* ‘on account of, because of’, for instance, has its origin in a form related to Pth. *wsn’d* /wasnāδ/, MP *wšn* /wašn/, OP *vašnā* ‘by grace of’, although it formally corresponds to none of the above fully; SZEMERÉNYI (1966) suggests that the original Parthian form *wasn resulted from a corruption with Pth. *r’d* /rāδ/ ‘for, owing to’. For numbers, the evidence is clearer: Arm. *hazar* ‘1,000’ < WMIr. *hz’r* /hazār/ and Arm. *biwr* ‘10,000’ < WMIr. *bywr* /bēwar/ are exact matches.

Since HÜBSCHMANN’S (1897) work, there has as yet been no comprehensive study or lexicon that discusses the entirety of Parthian or generally Iranian lexical items borrowed into Armenian; some efforts in this direction have been made in SCHMITT (1983), which contains a thematically grouped list of some common loanwords, and in GIPPERT (1993), discussing Iranian loans common to Georgian and Armenian at some length; MARTIROSYAN (2021) discusses Iranian personal names in Armenian. Instead, research in the latter half of the 20th century focused on supplementing, specifying, or correcting the compendium provided by HÜBSCHMANN in smaller contributions, often etymological studies of a small set of words (cf., e.g., PÉRIKHANIAN 1982; CONSIDINE 1984; BAILEY 1989 on Arm. *napastak* ‘hare’).

2.2.4 Derivational morphology

Armenian derivational morphology has also adopted a significant number of affixes of Parthian origin, which are used productively from the beginning of literary attestation onwards.

Some of them derive from original Parthian affixes, e.g. Arm. *-ik* < Ir. *-ika- as in Arm. *spasik* ‘servant’, *martik* ‘fighter, combatant’; Arm. *apa-* < Ir. *apa- / *upa- as in Arm. *aparank* ‘palace; house’, *apagovem* ‘to blame’.

Other common suffixes are based on Iranian compounds; these include, amongst others, Arm. *-kert* < Ir. *-kṛta-, cp. Arm. *dastakert* ‘building, village’, Pth. *yzygyrd* /yazdegird/ ‘divine, made by the gods’; Arm. *-stan* < Ir. *-stāna-, cp. Arm. *aspastan* ‘(horse) stable’, Pth. *d’dyst’n* /dādestān/ ‘id.’.

Both types of suffixes are productive both with inherited and borrowed lexical material.

2.2.5 Phraseology

Next to direct loans and grammaticalised suffixes, Armenian developed (semi-)calques on the basis of Parthian models, including both nouns and complex predicates.

Some examples include: Arm. *barekam* (< *bari-a-kam) ‘friend’, cp. Pth. *šyrg’mg* /šīrgāmag/, with an original meaning ‘well-wisher’ (cp. MP/Pth. *k’m* ‘wish, desire’); Arm. *jerbakal* ‘prisoner (lit. taken by the hand)’, cp. Pth. *dstgrb* /dastgraß/, with an exact correspondence of *jerb-* (cp. *jeṛn* ‘hand’) and *dst* as well as *-kal* (cp. Arm. *kalay*, suppletive aorist of *unim* ‘to have, hold’) and *-grb* (cp. Pth. *gyrw-* / *gīrw* ‘to take, seize’); Arm. *heṛi arnel* ‘to make remote, remove’, cp. Pth. *dūr kar-*, NP *dūr kardan*.

The Armenian reflexive *anjn iwr* ‘himself (lit. his own soul)’ is modelled on an Iranian original as well; cp. WMIr. *xwys gryw* /xwēš grīw/ and *xwys tn* /xwēš tan/ ‘own body’, YAv. *huua- tanu-* (cp. MEYER 2013).

2.3 Middle Persian

Direct, extended contact between Armenian and Middle Persian, going beyond interactions resulting from trade relationships, are likely to have commenced only with the fall of the Parthian Empire in 224

CE and the take-over of the Sasanians, but most probably only began properly with the integration of Armenia into the Sasanian Empire as a *marzpanate* in 428 CE.

Loanwords from Middle Persian are therefore much more restricted in scope, pertaining only to a limited number of lexical fields; between Parthian and Middle Persian loans there is, as Bolognesi (1980:33) summarises,

“una differenza qualitativa fondamentale ...: [i prestiti medio-persiani] sono essenzialmente costituiti da nomi di professioni e di monete, titoli onorifici, nomi di mesi, di luoghi e di persone, termini tecnici insomma”.

They do not show the same categorical diffusion as their Parthian counterparts in only yielding nominal loans. Most borrowings are found in military and administrative terminology; examples of these include: *šah* and *šahanšah* ‘(high) king’ < MP *š’h* /šāh/; Arm. *marzpan* ‘governor’ < MP *mrzb’n* /marzbān/; Arm. *payman* ‘condition, state’ < MP *pym’n* /paymān/; Arm. *p’uštīpan* ‘defender, bodyguard’ < MP *pwštīb’n* /puštībān/ (cp. an earlier loan, potentially from Parthian, Arm. *paštpan*); Arm. *tohm* ‘family, tribe’ < MP *twhm* /tōhm/ (cp. Pth. *twxm* /tōxm/).

Middle Persian loans can further be differentiated from those of Parthian origin by means of diverging phonological developments, which are particular to South-West Iranian languages (cf. Table 2).

SWIr. (MP)	NWIr. (Pth.)	Arm.
<i>h</i>	<i>s</i>	<i>akah</i> ‘knowledgeable’ < MP <i>’g’h</i> /āgāh/ (Pth. <i>’gs</i> /āgas/)
<i>d</i>	<i>z</i>	<i>dastak</i> ‘wrist, palm’, cp. MP <i>dst</i> /dast/ (Av. <i>zasta-</i>)
<i>y</i>	<i>δ</i>	<i>spayapet</i> ‘general-in-chief’, cp. Arm. <i>sparapet</i> (Av. <i>spāda-</i> ‘army’)
<i>l</i>	<i>rd</i>	<i>sałar</i> ‘leader’ < MP <i>s’l’r</i> /sālār/ (cp. Pth. <i>srd’r</i> /sardār/)
<i>d</i>	<i>b</i>	<i>darapan</i> ‘porter’, cp. MP <i>dr</i> /dar/ (cp. Arm. <i>barapan</i> < Pth. <i>brb’n</i> /barbān/)
<i>s</i>	<i>sp</i>	<i>sah</i> ‘army’, cp. Av. <i>spāda-</i>

Table 2: Different consonantal reflexes of Parthian and Middle Persian

There are also numerous *Doppelentlehnungen*, viz. words borrowed from Parthian and Middle Persian at different stages, e.g. Arm. *mogpet* (< Pth. *mgpyt* /maybed/) and *movpet* (< MP).

The limited occurrence of Middle Persian loans in Armenian, their categorical restriction to nouns, and their unproductivity are a reflection of the far more basic level of language contact that existed between the two languages, particularly as contrasted with Parthian.

2.4 Other Iranian influence

2.4.1 East Iranian

Some East Iranian loans, which find correspondences most closely to forms otherwise almost exclusively attested in Sogdian, have made their way into Armenian: Arm. *kari* ‘very’ < Sogd. *k’δy* /kāḍi/; Arm. *baw* ‘enough’, cp. Sogd. *β’w* /βāw/ ‘satiety’. More recent suggestions for East Iranian loans into Armenian, e.g. Arm. *zawr* ‘troops, army’ < Sogd. *z’wr* /zāwar/, are unlikely to withstand closer scrutiny, since the forms in question are attested as such also in Parthian.

Since direct interactions between Armenians and Sogdians seem unlikely, the *communis opinio* is still based on HENNING’S (1958) proposal that the East Iranian material is of Parnian origin. The Parnians, whose language is otherwise unattested, are said to have conquered the Parthians in about 240

BCE, and then to have adopted their language. Remnants of Parnian vocabulary would thus have found their way into Armenian by mediation of Parthian, in which some Eastern Iranian material is attested, e.g. Pth. *hnd* /hand/ ‘blind’ (contrast MP *kwr* /kōr/ ‘id.’).

2.4.2 A third West Middle Iranian language?

Similar to the set of words attributed to Eastern Iranian, certain loanwords from the Middle Iranian period suggest that there may well have been a third West Middle Iranian dialect.

On a phonological level, this assumption is based on the development of Clr. **d* > *h* in a small number of Armenian borrowings; these include Arm. *zrahk* ‘cuirass’, cp. YAv. *zrāda-* (contrast the Aramaic loanword *zrd* /zardā/), which also occurs in Zoroastrian Middle Persian as *zryh* /zrēh/; Arm. *srah* ‘hall, courtyard’, cp. Clr. **srāda-*, and Zoroastrian MP *sr’d*, *sr’y*.

Furthermore, recent forays into this field suggest that the adjectival suffix Arm. *-agin* ‘like X, endowed with X’ was derived from such a third language; a similar suffix, Arm. *-kēn* < Pth. *-gyn* /-gen/, cp. Clr. **-k-aina* (Av. *zaran-aēna* ‘golden’, OP *aθang-aina* ‘from stone’), is also attested and illustrates the phonological difficulties of deriving Arm. *-agin* from its known Middle Iranian counterparts. KORN AND OLSEN (2012) and KORN (2013) (on a different note) therefore suggest an origin in a third language (pace SCHMITT 2001:85). Yet, there is a caveat: borrowing of derivational morphology usually requires prolonged and extensive language contact, such as in the case of Parthian; it is unclear how the third language with its few identifiable loans fits into this picture.

Since this hypothesised dialect has left few traces beyond what has been mentioned above, it cannot be securely identified; a connection to the equally poorly attested Median language as suggested by PÉRIKHANIAN (1966), e.g. as Middle Median, is not to be excluded *a priori*, but cannot be verified.

3 Replication of smaller patterns

Beyond lexical, morphological, and phraseological loans, recent work suggests that Armenian has further undergone syntactic influence from the West Middle Iranian languages. Such influence is expressed in the replication of specific functions of a model language word or syntagma and its mapping on a suitable word or syntagma in the target language (‘pivot matching’).

In all instances, illustrative examples are provided without any further discussion of borrowing processes; the latter will be addressed in §5 below. Equally, it cannot be excluded that the parallels described have arisen independently in West Middle Iranian and Armenian; their relative peculiarity suggests, however, that this explanation is less likely. For details, see MEYER (2023:ch. 6).

3.1 Nominal relative clauses (?)

The West Middle Iranian languages have developed a nominal relative construction, also called *ezāfe*, in which determinand and determinans are connected by means of a relative particle.

- (1) *tōhm čē amāh*
 family EZF 1.PL.POSS
 ‘our family’ (NPi pa §65)

Some have argued (e.g. BENVENISTE 1964:35) that Armenian has replicated this structure in its nominal relative clauses.

- (2) *anun Astucoy or tearn=n araracoc'*
 name.NOM.SG God.GEN.SG REL/EZF.NOM.SG lord.GEN.SG=DET creation.GEN.PL
 'the name of God, the Lord of [all] Creation' (Eznik IV.1)

As yet, however, this analysis is uncertain for at least two reasons: not all Armenian sentences show finite verbs, esp. the copula (cf. §4); nominal relative clauses are attested in other Indo-European languages (cf., e.g., PROBERT 2015:407–414 on Greek; SEILER 1960 on Avestan).

The *ezāfe* construction is at this stage not obligatory in any of the languages in question, and not particularly frequent in Armenian. Results of a corpus study may provide further insight.

3.2 Intensifier-cum-anaphora

Both Parthian and Middle Persian share a function word Pth. *wxd* /*wxad*/, MP *xwd* /*xwad*/ < InIr. **h̥uatah* which serves three distinct purposes: intensifier, anaphora, and switch-function marker (cf. MEYER 2013.)

- (3) 'w=š yyšw' w'(x)t lw nxw(š)[t] tw wxd w'xt kw
 and=3.SG.OBL Jesus speak.3.SG.PRS COMP first.SG 2.SG INT speak.3.SG.PRS COMP
 'z hym
 1.SG.DIR be.1.SG.PRS

'And Jesus said to him: "You yourself said that I am ..."' (MKG 1193; Pth.)

- (4) 'wt pt r'h kw šwyd tgnbnd 'w dšt 'yw wzrg w wy'b'n
 and on path when go.3SG.PRS quickly to steppe; one great and desolate
 y'dyndyh 'w_y škyft grm 'hyn(dy)h 'wt wxd wgw d wšynd 'wt
 reach.3SG.OPT; and very hot be.3SG.OPT; and ANAPH; starve.PTCP hungry and
 tšynd 'hyndyh
 thirsty be.3SG.OPT;

'And on the path, as he walks, he quickly reaches a great and desolate steppe. And it was very hot, and he was starving, hungry and thirsty.' (MKG 797; Pth.)

- (5) wd jnyd ('w h)[w] d'lw g mrnyh cy wxd 's(t) [xyn
 and fell.3.SG.PRS OBJ DEM.SG tree.SG; deadly.SG; REL SWITCH-FCT; be.3.SG.PRS hatred.SG
 'And he fells the Tree of Death, which itself is hatred.' (LN 27; Pth.)

An etymologically related but non-identical word is found in Arm. *ink'n* < **en-suom*, which shows a similar functional distribution. Switch-function marking in Armenian is less common, likely owing to its more developed case morphology.

- (6) kam orpes z=noun ink'n ašxarhi hoviw kac'uc'eal,
 or how OBJ=same.ACC.SG int land.GEN.SG shepherd.ACC.SG appoint.CVB
 vaye lēin i norun vardapeteut'ean=n
 enjoy.3PL.PST in same.GEN.SG teaching.LOC.SG=DET
 'Or how they appointed the very same man as shepherd of the land, and enjoyed his teaching.'
 (Ag. §14)

- (7) ew ibrew emut i nawn, gnac'in zkni nora ašakert-k'=n.
 and when enter.3.SG.AOR in ship.ACC.SG go.3PL.AOR after DEM.GEN.SG disciple-PL=DET
 ew aha šaržumn mec elew i covun ... ew ink'n
 and behold earthquake.NOM.SG big become.3SG.AOR in sea.LOC.SG and ANAPHORA

nnjēr

sleep.3.SG.PST

‘And when he embarked upon the ship, his disciples followed him. And behold, there was a great earthquake in the sea ... and he was asleep.’ (Mt. 8:24)

- (8) ... *ew yišel* *z=ban=n* *t(eař)n* *y(isu)si* *z=or*
and remember.INF.PRS OBJ=word.ACC.SG.=DET lord.GEN.SG Jesus.GEN.SG OBJ=REL
ink‘n isk asac‘
SWITCH-FCT INT say.3.SG.AOR
‘... and to remember the words of the Lord Jesus, which indeed he said.’ (Act. 20:35)

It must be noted that Arm. *ink‘n* is an inflecting pronoun as opposed to its Iranian counterparts, and that it is also used as a reflexive pronoun. It is difficult to determine the relationship between the pronoun’s various functions; it appears likely, however, that those not paralleled in Iranian are ‘native’ to Armenian.

3.3 Complementiser

The Armenian complementiser (*e*)*t‘ē* is used in a number of contexts (for an overview, see JENSEN 1931). For the present purpose, its most pertinent functions are the introduction of CPs, direct, and indirect speech.

- (9) *teseal t‘ē gelec‘ik ē ašxarh=s*
see.CVB COMP good be.3SG.PRS world=DET
‘... on seeing that the world is good’ (Ezn. 244)
- (10) *asen t‘ē y=aysm vayrē ok‘ andr mi žlesc‘i*
say.3PL.PRS COMP from=DEM.ABL.SG place.ABL.SG INDF onward NEG dare.3.SG.AOR.SBJV
ert‘al
go.INF
‘And they say: from this place no one shall dare to go forth.’ (Ezn. 219)

It is particularly worth noting the that in its function as indirect speech marker, (*e*)*t‘ē* also introduces indirect *wh*-questions.

- (11) *zarmac‘an ew asen t‘ē ziard valvalaki c‘amak‘ec‘aw t‘zeni=n*
be-astonished.3.PL.AOR and say.3.PL.PRS COMP how suddenly wilt.3SG.AOR fig-tree=DET
‘They were astonished and said: “Why has the fig-tree suddenly wilted?”’ (Mt. 21:20)

The same combination of functions in Parthian is expressed by the function word *kw* /kū/. Here, too, even *wh*-questions are introduced by COMP.

- (12) *hō=mān ōh kāmēd kū hō ranž ud aβyām*
DEM=1PL thus wish.ptcp COMP DEM toil and pain
‘This we wish, that toil and pain ...’ (NPi pa §72)
- (13) *pad wuzurg šādift ō man wāxt kū až kū ay tō,*
with great joy to 1.SG.OBL say.PTCP COMP from where be.2.SG.PRS 2.SG.DIR,
man bay ud anjīwag?
1.POSS god and saviour
‘And [she] said to me with great joy: “From where do you come, my Lord and Saviour?”’ (MKG 126–128)

As with *ink'n* above, so *(e)t'ē* too has further functions not paralleled in West Middle Iranian, e.g. introducing conditional clauses (MEYER (in press*b*); Pth. uses 'g /ag/, cf. DURKIN-MEISTERERNST 2014:444).

4 The Armenian periphrastic perfect

Previous explanations of this construction have been proposed by, amongst others, BENVENISTE (1959); STEMPEL (1983); WEITENBERG (1986); for reasons of time and space, they will not be discussed here. Suffice it to say that each proposal has flaws (some irredeemable) which occasioned the next attempt; the current approach may well be no different. For a succinct review of these, MEYER (cf. 2022*b*).

4.1 Outline

Traditionally, the Armenian periphrastic perfect is said to be made up of a past participle in *-eal* and an non-obligatory form of the copula.

The participle on its own, when used in its adjectival function, may be intransitive or passive in meaning; this is reflected, e.g., in the word for 'apostle', *arāk'eal* (send.PTCP). A transitive participle on its own is almost unattested.

- (14) *z=amēn mnac'eals y=anjn arnēr*
 OBJ=all remain.PTCP.ACC.PL to=soul.ACC.SG take.3.SG.PST
 'And he took unto himself all [those who] remained' (Koriwn §26)

In the periphrastic perfect, the subject in passive and intransitive predicates shows NOM marking, and the copula (if present) agrees with the subject in number and person. An optional passive agent is expressed by a PP with Arm. *i* + AB.

- (15) *i galstenē kenarari ordwoyn Astucoy bac'eal ē duṛn*
 by arrival.ABL.SG life-giving son.GEN.SG god.GEN.SG open.PTCP be.3SG.PRS door.nom.sg
šnorhi olormut'ean
 grace.GEN.SG mercy.GEN.SG
 'The door to the grace of mercy was opened by the arrival of the life-giving son of god' (ŁP II.12)

- (16) *ew ibrew ayl ok' oč' ēr mnac'eal y=azgēn*
 and when other INDF NEG be.3.SG.PST remain.PTCP from=clan.ABL.SG
 'And when no one else from this clan (had) remained, ...' (PB III.9)

In transitive predicates, the agent shows GEN, the object ACC morphology, while the copula only occurs in a fixed 3.SG form without agreeing with either agent or object.

- (17) *bazum anhamar č'aris gorceal ē jer*
 many innumerable evil.ACC.PL do.PTCP be.3SG.PRS 2.PL.GEN
 'You have committed innumerably many evils.' (ŁP III.36)

Armenian does not morphologically distinguish NOM and ACC in the singular, but has an OBJ-marking *nota accusativi* Arm. *z=*, which may reflect definiteness/specificity-based differential object marking (cp. SCALA 2011).

- (18) *ew t'agaworin patasxani tueal asē*
 and king.GEN.SG answer.NOM/ACC.SG give.PTCP say.3SG.PRS
 'And the king gave an answer and said: ...' (PB III.35)

By the 8th century CE, NOM had encroached on the domain of GEN agents in transitive verbs almost completely. This is already foreshadowed sporadically in earlier texts. Equally, GEN in a few instances occurs as an intransitive subject, suggesting that already by the end of the 5th century, there may have been some confusion as to the correct usage of both cases.

- (19) ... *z=or ēr gorceal nengawor išxan=n Siwneac'*
 OBJ=REL.ACC.SG be.3.SG.PRS do.PTCP treacherous prince.NOM.SG=DET Siwnik'.GEN.PL
Vasak ew ayl naxararac'=n
 PN.NOM.SG and other nobleman.GEN.PL=DET
 '[the terrible deed] which Vasak, the treacherous prince of the Siwnik', and the other noblemen had done' (LP III.17)

- (20) *yaynžam i mi vayr žołoveal naxararac'=n amenec'un*
 then in one place.ACC.SG assemble.PTCP nobleman.GEN.PL=DET all.GEN.PL
 'At that time all noblemen assembled in one place.' (LP III.8)

Since Armenian is a pro-drop language, and owing to frequent object ellipsis and lack of finite verbs or other agreement features, predicates can often only be securely construed in context.

Next to its role as a full verb, the participle alone occurs frequently as a converb, which may for present purposes be defined as a non-finite verb which syntactically depends on another verb form, but is not its argument; rather it may be an adjunct (cf. NEDJALKOV 1995:97).

In converbial function, expressing an action subordinate to the main verb's, the participle may share the main verb's subject/agent or supply its own; even without express subject/agent, converb and main verb subject/agent may differ owing to pro-drop.

- (21) *ew merkac'uc'eal dahčac'=n z=surb=n asac' eric's*
 and undress.CVB executioner.GEN.PL=DET OBJ=holy.ACC.SG=DET say.3.SG.AOR thrice
 'And when the executioners had undressed the saint, [he] said three times: ...' (LP III.38)

Although intransitive subject and transitive agent show different morphological marking, Armenian appears to operate largely on a notion of subjecthood that is case-independent. Accordingly, co-ordinated transitive and intransitive predicates can share a subject/agent in either case. Which case is chosen appears to be determined by either proximity of verb and subject/agent, or the question which is the main action.

- (22) *ew lueal z=ays amenayn bans anawrinin Mihrnersehi ew xndalic'*
 and hear.PTCP OBJ=DEM all word.ACC.PL wicked.GEN.SG PN.GEN.SG and joyful
eleal
 become.PTCP
 'And when the wicked Mihrnerseh heard all these words, he rejoiced' (LP III.2)

Note that the converbial usage is the most frequent at c. 54%.

4.2 The West Middle Iranian past tense

While Armenian has two further synthetic past tenses (imperfect, aorist), both West Middle Iranian languages only feature a periphrastic past tense, composed of the past participle and a copula.

As in Armenian, the participle can be used adjectivally (cf. DURKIN-MEISTERERNST 2014:253). Intransitive and passive predicates require direct case subjects; the copula agrees with the subject. Transitive predicates, on the other hand, show oblique case agents and direct case objects; the copula agrees with the object (ergative alignment). Note, however, that a clear differentiation in case morphology is only possible in personal pronouns (e.g. 1.SG.DIR *az*, 1.SG.OBL *man*); ergativity is otherwise only noticeable through verbal agreement and in oblique enclitics serving as agents.

- (23) *'wd 'z 'gd hym kw 'c bzkr bwj-'n*
and 1.SG.DIR come.PTCP be.1SG.PRS CONJ from evil-doer rescue-1.SG.SUBJ
'And I have come so that I may rescue (you) from the evil-doer.' (R/VI/64a)

- (24) *'wd pd tw bst dydym 'w hrwyn dwšmn-yn*
and by 2.SG bind.PTCP diadem for all enemy-PL
'And a diadem was bound by you for all enemies.' (AR/VI/56a)

- (25) *byc 'w's cy=m dyd 'yy 'w=m tw sxwn 'šnwd*
but now COMP=1SG see.PTCP be.2SG.PRS CONJ=1SG 2.SG speech hear.PTCP
'But now that I have seen you and heard your speech ...' (MKG 1398–1400)

The West Middle Iranian ergative construction in the past is replaced by a NOM–ACC construction by the 9th century.

4.2.1 Differences with the Armenian periphrastic perfect

There are three key differences between the Armenian and West Middle Iranian construction:

1. The West Middle Iranian past tense never shows a 3.SG copula, and shows object agreement in all other tenses.
2. The object in Parthian / Middle Persian is marked as DIR, whereas Armenian has ACC.
3. The agent in West Middle Iranian is OBL, while Armenian shows GEN.

Ad (1): the fixed 3.SG copula in Armenian is rare to begin with and gains currency over the course of the 5th century; it is therefore a relatively recent 'invention'. The pluperfect in both Armenian and West Middle Iranian DOES have a 3.SG copula, however:

- (26) *ew Astucoy z=mez i jeṛanē č' =ēr toleal*
and god.GEN.SG OBJ=1PL.ACC from hand.ABL.SG NEG=release.PTCP
'... and god had not released us from his hand.'

- (27) *ō abarēn nē paryābād ud až padrāst būd ahāz*
toward higher NEG achieve.PTCP and from prepare.ptcp become.ptcp be.3SG.PST
paryūdag
lose.PTCP
'[The demon] did not reach the upstairs, and lost what he had prepared.' (M741/R/6ab)

Note that other split-ergative languages have developed fixed 3.SG copula forms (e.g. Talyši, cf. PAYNE 1979:442); this appears to be a step towards or sign of de-ergativisation.

Ad (2): it is conceivable that the original pattern may have involved a NOM object, which owing to NOM/ACC.SG homophony was re-analysed as ACC. Also note that West Middle Iranian also occasionally uses an object marker 'w /ō/ to mark the DIR-case object.

Ad (3): it is unclear by what time the West Middle Iranian case system had eroded; according to SKJÆRVØ (1983), a differentiation in plural nouns exists up to the 3rd century CE. The Armenian choice of the genitive may thus be based on functional parallels (possession marking, use after prepositions, etc.).

5 Analysis

The parallels between the West Middle Iranian, specifically Parthian, past tense construction and its Armenian counterpart suggest, despite the differences outlined, that the former may have influenced the development of the latter.

Pattern replication, according to Ross (2007), rarely consists of the wholesale adoption of a model language pattern, but rather an adaptation thereof within the grammar of the target language by pivot matching. Owing to the lack of Armenian data prior to Iranian influence, it is impossible to tell which aspects of the Armenian construction are original or 'native' and which have been replicated in contact. In the 'mental grammar' of a bilingual speaker of Armenian and Parthian, however, the two constructions are sufficiently overlapping (e.g. as regards the use of the participle), the various cases involved, and the tense/aspect reference to allow for pivot matching and polysemy copying (in the terms of MATRAS AND SAKEL 2007).

The evidence from lexicon, phonology, morphology, etc. unequivocally points to an intense and long-lasting contact situation between Armenian and the Iranian languages, but cannot paint a clearer picture as to under which circumstances it occurred. There are further neither documents containing traces of code-switching or code-mixing, as in some other instances of historical language contact (e.g. Norman French in Britain) which might clarify the situation, nor is there any suggestion of loans from Armenian into Parthian.

Certain historical facts, as represented in historiographic literature, may provide at least an inkling of an idea, however:

1. Iranian-Armenian intermarriage was not uncommon;
2. nor was the exchange of wards between Armenian and Iranian noble families;
3. the fall of the Parthian Empire in 224 CE and subsequently frequent wars between Partho-Armenian nobility and their Sasanian neighbours;
4. the Christianisation of Armenian and Parthian speakers in Armenia at the beginning of the 4th century;
5. complete lack of evidence of Arsacid Parthian inscriptions on core Armenian territory;
6. the lack of new Parthian language material in Armenia or Persia from the end of the 3rd century CE onwards.

The intermixing between Parthian and Armenian nobility is likely to have led to some degree of bilingualism in that class, which the linguistic data supports. The historical data may be taken to tentatively suggest the possibility of a ‘superstrate shift’: the Parthian ruling class, over the course of generations, adopted Armenian as their primary language; over the course of time, Parthian was relegated to a secondary language and may have died out. The varying degrees of bilingualism, both in primary Parthian and Armenian speakers, may then have led to the replication of the patterns discussed (cf. Figure 3 and MEYER 2022a).

Inevitably, this analysis remains speculative; the kind of data, linguistic or otherwise, necessary to prove the above does not exist. Nonetheless, this account can address the great formal variety of the Armenian periphrastic perfect and some other constructions more succinctly than prior attempts.

In this vein, it is worth keeping in mind THOMASON’s tenet that ‘anything goes in language contact’ (2008; 2010).

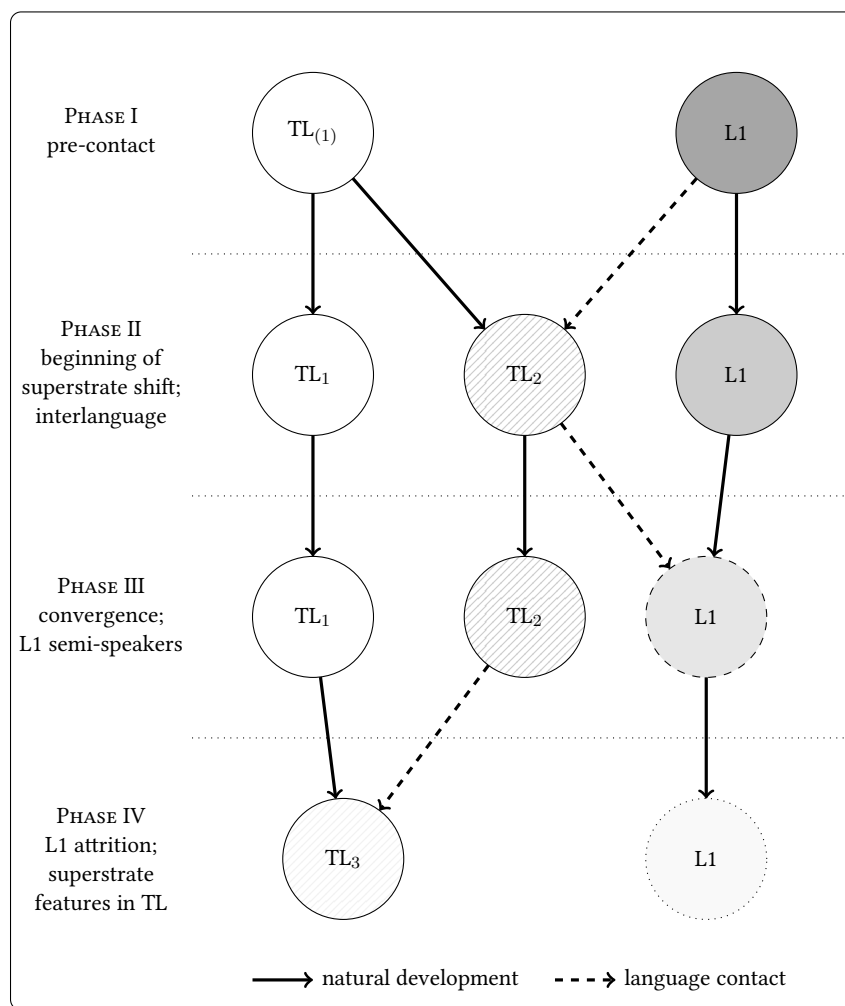


Figure 3: Simplified Language Shift Schema: Language shift occurs in distinct phases. Phase I represents the languages in their unaltered, pre-contact state. In phase II, L1-speakers learn the TL, and through imperfect learning and/or pattern replication create an interlanguage TL₂; their L1 remains constant. Phase III sees the rise of importance of TL₂ for L1-speakers, and not all speakers acquire L1 fully or without interference; L1 converges towards TL₂. In phase IV, finally, TL₁ and TL₂ merge into TL₃ as former L1-speakers integrate fully into the TL society; L1 may live on as a heritage language, or disappear entirely.

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